Randy Crawford
Manager
Regulatory Assurance
225-381-4177

RBG-48262

10 CFR 50.73

November 15, 2023

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2023-003-01, Drywell Atmosphere Radiation Monitor Particulate and Gaseous Channels Inoperable due to Out of Tolerance Conversion Factors

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

On September 14, 2023, Licensee Event Report 50-458 / 2023-003-00, Drywell Atmosphere Radiation Monitor Particulate and Gaseous Channels Inoperable due to Out of Tolerance Conversion Factors (ML23257A257) was submitted by Entergy Nuclear Operating Inc. This letter is a supplement to the report to provide the updated results of the causal evaluation. The updated information is denoted by revision bars located in the right-hand margin. The report is submitted in accordance with 10 CFR 50.73.

This document contains no commitments.

Should you have any questions, please contact Mr. Randy Crawford, Regulatory Assurance Manager, at 225-381-4177.

Respectfully,

Randy Crawford

RC/twf

Enclosure: Licensee Event Report 50-458 / 2023-003-01, Drywell Atmosphere Radiation Monitor Particulate and Gaseous Channels Inoperable due to Out of Tolerance Conversion Factors

cc: NRC Region IV Regional Administrator - Region IV
NRC Senior Resident Inspector - River Bend Station

Enclosure

RBG-48262

Licensee Event Report 50-458 / 2023-003-01, Drywell Atmosphere Radiation Monitor Particulate and Gaseous Channels Inoperable due to Out of Tolerance Conversion Factors

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2024

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
River Bend Station, Unit 1	■ 050 / ☐ 052	458	1 OF 3

4. Title
Drywell Atmosphere Radiation Monitor Particulate and Gaseous Channels Inoperable due to Out of Tolerance Conversion Factors

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
07	17	2023	2023	- 003 -	01	11	15	2023	N/A	☐ 050 N/A
									N/A	☐ 052 N/A

9. Operating Mode	10. Power Level
1	100%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20		10 CFR Part 50			
	☐ 20.2203(a)(2)(vi)		☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☑ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Randy Crawford, Manager – Regulatory Assurance	225-381-4177

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
D	IL	MON	G063	Y	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)		N/A	N/A	N/A

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced type written lines)

On August 29, 2023, with River Bend Station (RBS) operating at 100% power, personnel recognized that the Drywell Atmosphere Radiation Monitor (RMS-RE112) particulate and gaseous channel conversion factors were out of tolerance. The monitor was declared inoperable and the appropriate Technical Specification (TS) Condition was entered.

This event was caused by configuration controls that were inadequate to prevent human performance errors in development of radiation detector digital parameters and transfer of parameters into the detector's database at initial setup.

The RMS-RE112 conversion factors have been corrected and the monitor was declared operable on September 7, 2023.

This condition is being reported as an Operation or Condition Prohibited by Technical Specifications per 10 CFR 50.73(a)(2)(i)(B).

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
River Bend Station, Unit 1	■ 050 / ☐ 052	458	2023	- 003	- 01

NARRATIVE

EVENT DESCRIPTION

On August 29, 2023, River Bend Station (RBS) personnel recognized that the Drywell Atmosphere Radiation Monitor (RMS-RE112) [IL] [MON] particulate and gaseous channel conversion factors were out of tolerance. The monitor was declared inoperable and Technical Specification (TS) Limiting Condition of Operation 3.4.7, Reactor Coolant System Leakage Detection Instrumentation, Condition B was entered. The RMS-RE112 conversion factors were corrected, and the monitor was declared operable on September 7, 2023.

The discrepancy was identified as part of an extent of condition review performed due to the Notice of Violation (ML23201A132) issued to RBS on August 15, 2023, for failing to maintain reliable and accurate indications on five effluent radiation monitors.

On July 17, 2023, a condition report was initiated requesting a work order to review and validate the RMS-RE112 conversion factors. On August 29, 2023, RBS personnel recognized that information had been available to determine that the conversion factors were out of tolerance since the condition report was initiated on July 17. Therefore, the date of discovery is July 17, 2023.

The conversion factors have been out of tolerance for the life of the plant. Therefore, this event is being reported as an Operation or Condition Prohibited by Technical Specifications per 10 CFR 50.73(a)(2)(i)(B).

EVENT CAUSE

This event was caused by configuration controls that were inadequate to prevent human performance errors in development of radiation detector digital parameters and transfer of parameters into the detector's database at initial setup.

SAFETY ASSESSMENT

Limits on leakage from the Reactor Coolant Pressure Boundary (RCPB) are required so that appropriate action can be taken before the integrity of the RCPB is impaired. Leakage detection systems for the Reactor Coolant System (RCS) are provided to alert operators when leakage rates above normal background levels are detected and to supply quantitative measurement of rates. RMS-RE112 comprises one of three RCS leakage detection instrumentation systems required by RBS TS 3.4.7. At least one of the two remaining RCS leakage detection instrumentation systems of TS 3.4.7 remained operable during the 3-year LER past operability period.

RMS-RE112 is not used to recognize and properly classify emergency conditions and is not classified as Equipment Important to Emergency Response (EITER).

This event was of minimal significance to the health and safety of the public.

CORRECTIVE ACTIONS

Completed:

The RMS-RE112 conversion factors were corrected and the monitor was declared operable on September 7, 2023.

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			YEAR	SEQUENTIAL NUMBER	REV NO.
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NARRATIVE

Procedures have been implemented that reduce the chance of human performance errors during development of radiation detector digital parameters and transfer of parameters into the detector's database at initial setup.

PREVIOUS SIMILAR OCCURRENCES

No previous similar occurrences identified.

Energy Industry Identification System (EIIS) codes are identified in the text as [XX]. River Bend equipment codes are identified as (XX).